

Instructions:

Your task is to analyze the **2023 MGM Resorts cyberattack** using the incident analysis worksheet provided.

Your analysis must highlight **key lessons in cyber forensics and incident investigation**. Focus on the legal and ethical challenges associated with the **digital forensics' techniques** used in this case.

Your final submission should provide practical insights into the intersection of **cyber forensics**, law enforcement, and the **legal admissibility of digital evidence** based on this case study.

Submission Instructions :

- Complete the incident analysis worksheet.
- Export your completed work as a **PDF file**.
- Submit the PDF via myCourses.

If revisions are needed, you can upload a new version of your assignment. This will override the existing file, and only the most recent version will be saved.

Individual or Teamwork of 2		
Student(s) identification		
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6 Points	Title of Incident:	Provide a brief name for this incident 2023 MGM Resorts Cyberattack (Social Engineering and Ransomware Attack)
	Discovery date:	When was this incident discovered? The incident was publicly discovered and reported on September 10, 2023 , when MGM Resorts began experiencing major system outages across its casinos and hotels.
	Start date of Incident:	When did this incident begin? The attack likely began around September 9, 2023 , when attackers gained access to MGM's internal systems through social engineering techniques.
	End date of incident:	When did this incident end? The major operational disruptions lasted for about 10 days , with systems gradually returning to normal by late September 2023 .
	Organization affected:	What is the organization that was directly impacted? MGM Resorts International

		Number of employees	Approximate number: Approximately 80,000+ employees worldwide
		Revenues	Approximate revenues: Around \$16 billion annually
		Country	Where is the head office located? United States (Headquarters in Las Vegas, Nevada)
		Line of business	What is their main business: Hospitality, casino operations, hotels, and entertainment services.
4	Cause of the incident:	Accident, Error or Voluntary? The incident was caused by external attackers who intentionally targeted MGM Resorts. The attackers used social engineering techniques to manipulate employees into granting system access.	
	Aspect affected:	Confidentiality, Integrity or Availability? There could be multiple answers Confidentiality: Sensitive customer data such as driver's license numbers and personal information may have been exposed. Integrity: Internal systems were compromised by attackers who gained unauthorized access. Availability: Many MGM systems went offline, including hotel check-ins, slot machines, digital room keys, and payment systems.	
10	Incident description		
	Provide a paragraph describing the incident itself. Don't cover its resolution. The MGM Resorts cyberattack in September 2023 was one of the most disruptive cyber incidents in the hospitality industry. The attackers used social engineering techniques to impersonate an employee and convince the company's IT help desk to reset credentials. Once inside the system, they gained access to MGM's internal network and deployed ransomware tools. This caused multiple internal systems to shut down across MGM properties in Las Vegas and other locations. Customers experienced problems checking into hotels, accessing digital room keys, using ATMs, and even operating casino slot machines. The attackers reportedly attempted to extort money from MGM in exchange for restoring access and not releasing stolen data.		
10	Direct impacts		
	Provide a description of how the operations of the affected organization have been negatively impacted and explain why. The cyberattack caused major operational disruptions across MGM's casino and hotel properties. Many automated systems stopped working, forcing employees to handle processes manually. Hotel guests had difficulty checking in, making payments, and accessing their rooms. Some slot machines and casino systems also stopped functioning, which directly affected revenue. Because MGM relies heavily on digital systems for daily operations, the outage significantly slowed down services and negatively impacted the customer experience. In addition, the attack damaged the company's reputation and raised concerns about how well customer data was protected.		
5	Direct cost		
	Do we know anything about the direct costs of this incident? How has this money been spent? You can speculate or extrapolate on this if the information is not available. The attack reportedly cost MGM Resorts over \$100 million in damages. These costs came from: 1. Lost revenue from casino and hotel operations 2. Incident response and cybersecurity investigations 3. System recovery and rebuilding infrastructure 4. Legal costs and regulatory compliance		

	5. Customer notifications and potential lawsuits These costs do not include possible long term reputational damage or future security investments.
10	Incident timeline
	Provide a timeline of the major events related to this incident. This should include at least the beginning of the incident, its discovery, and its resolution. September 9, 2023; Attackers gain access through social engineering targeting the IT help desk. September 10, 2023; MGM detects system disruptions and begins shutting down systems to contain the attack. September 11-14, 2023; Major service outages occur across MGM casinos and hotels. Customers report problems with hotel check-ins, digital keys, ATMs, and slot machines. Mid-September 2023; MGM begins restoring systems while working with cybersecurity experts and law enforcement. Late September 2023; Most systems return to normal operation.
5	Perpetrator
	Do we know who is the likely perpetrator of this incident? What is our level of confidence? Why? The attack is believed to have been carried out by the hacking group Scattered Spider, which has connections to the ALPHV/Black Cat ransomware group. Security researchers and law enforcement identified patterns in the attack methods that match previous attacks conducted by these groups. The same group also targeted Caesars Entertainment around the same time.
5	Motivation
	Do we know anything about the likely motive for causing this incident? What is our level of confidence? Why? The primary motivation was financial gain. Ransomware groups typically break into company's systems to steal data and demand a payment in exchange for not releasing the data publicly. These attacks are highly profitable for cybercriminal organizations. Because MGM refused to pay the ransom, the attackers caused significant disruption to pressure the company.
10	Vulnerabilities being exploited and techniques used by the attackers
	An incident can only happen if the organization was vulnerable to something. What exploited vulnerabilities led to this incident? What specific techniques were used by the attackers? You can speculate or extrapolate on this if the information is not available. The attackers mainly exploited human vulnerabilities rather than technical ones. They used social engineering to trick MGM's IT help desk into giving them access to employee credentials. Once inside the system, they were able to move laterally across the network. Techniques likely used included: 1. Social engineering and impersonation 2. Credential theft 3. Privilege escalation 4. Ransomware deployment 5. Network lateral movement From a cyber forensics perspective, investigators likely relied on log analysis, network monitoring, and digital evidence from compromised systems to understand how the attackers moved through the network.

10	Incident resolution
	What was done by the affected organization to resolve this incident and get back to normal operations? Can we say that the incident has been fully resolved or are there aspects that can never be fully resolved? MGM responded by shutting down many internal systems to stop the spread of the attack. They worked with cybersecurity firms and law enforcement agencies to investigate the breach and restore services. Systems were gradually rebuilt and brought back online. Although operations returned to normal, some consequences such as stolen personal data cannot be fully reversed. Data that was already accessed by attackers may continue to pose a risk for identity theft.
5	Communications to external parties
	Was the organization affected forthcoming on their communications? Did they seem to have a communication strategy? Did this have an impact on the incident? MGM did communicate publicly about the incident, but some critics argued that the information released was limited at first. The company eventually confirmed that a cybersecurity attack had occurred and warned customers that personal information may have been compromised. Their communication strategy appeared focused on controlling panic while restoring operations. However, earlier transparency might have helped customers better understand the risks and take precautions sooner.
10	Long-term impacts
	Has there been or do you anticipate any long-term impacts to the organization? Why? The long-term impacts could include: 1. Loss of customer trust 2. Potential lawsuits related to data breaches 3. Increased regulatory scrutiny 4. Higher cybersecurity spending 5. Stronger internal security policies The attack also serves as a warning to other organizations in the hospitality and gaming industries about the risks of social engineering attacks.
10	Lessons learned
	Could the organization have done anything differently to prevent this incident from happening? Can we say this organization had done their due diligence? One of the biggest lessons from the MGM cyberattack is that human error can be the weakest point in cybersecurity. Even large companies with advanced security systems can still be compromised if attackers successfully manipulate employees. The organization could have improved security by: 1. Implementing stronger identity verification for help desk requests 2. Providing better employee cybersecurity training 3. Using stricter multi-factor authentication controls 4. Monitoring suspicious account activity more closely From a cyber forensics and legal perspective, proper logging and evidence preservation are also critical. Digital evidence must be collected carefully to ensure it can be used in investigations and potentially in court.

	Although MGM had security measures in place, this incident shows that stronger preventive measures and employee awareness could have reduced the likelihood of the attack.
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References

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